

1 Solution — GIAM 1.1

Problem 5

1.1 Problem

Give a description of the set of rational numbers whose decimal expansions terminate. (Alternatively, you may think of their decimal expansions ending in an infinitely-long string of zeros.)

1.2 The Answer

A rational number a/b (in lowest terms, $b > 0$) has a *terminating* decimal expansion if and only if the only prime factors of b are **2** and **5**. Equivalently:

$$\left\{ \frac{a}{b} \in \mathbb{Q} : \gcd(a, b) = 1, b = 2^m \cdot 5^n \text{ for some } m, n \in \mathbb{Z}_{\geq 0} \right\}.$$

These rationals can also be characterised in several equivalent ways:

- **As integer-over-power-of-ten:** $\{ c / 10^k : c \in \mathbb{Z}, k \in \mathbb{Z}_{\geq 0} \}$.
- **As the ring $\mathbb{Z}[1/10]$:** the smallest subring of $\mathbb{Q}$ containing $\mathbb{Z}$ and $1/10$.
- **By decimal:** the rationals x with a finite-digit decimal representation $x = d_n d_{n-1} \cdots d_1 d_0 . d_{-1} d_{-2} \cdots d_{-k}$.

Call this set the **terminating decimals**, or sometimes the **decadic rationals** (or “dyadic-quintic rationals”).

1.3 Why it is True — Both Directions

$b = 2^m 5^n \implies a/b$ terminates.

Let $k = \max(m, n)$. Multiply both sides of a/b by $10^k/10^k$:

$$\frac{a}{b} = \frac{a \cdot (10^k/b)}{10^k}.$$

The ratio $10^k/b = 2^k 5^k / (2^m 5^n) = 2^{k-m} \cdot 5^{k-n}$ is an *integer* (because $k \geq m$ and $k \geq n$). So $a/b = c/10^k$ with $c = a \cdot (10^k/b) \in \mathbb{Z}$.

Now $c/10^k$ is — by definition of decimal notation — a decimal with the digits of c placed k positions after the decimal point. *Finite-digit*. So a/b terminates. ■

$$a/b \text{ terminates} \implies b = 2^m 5^n.$$

If a/b terminates after k decimal places, then a/b can be written as $c/10^k$ for some integer c and some $k \geq 0$ (just count the digits after the decimal point). So

$$\frac{a}{b} = \frac{c}{10^k} = \frac{c}{2^k \cdot 5^k}.$$

Cross-multiplying: $a \cdot 2^k \cdot 5^k = b \cdot c$, so $b \mid a \cdot 2^k \cdot 5^k$. Since $\gcd(a, b) = 1$ (lowest terms), b shares no prime factors with a , so $b \mid 2^k \cdot 5^k$. Hence every prime factor of b is in $\{2, 5\}$, i.e. $b = 2^m 5^n$ for some $m, n \geq 0$. ■

1.4 Visualising the Pattern

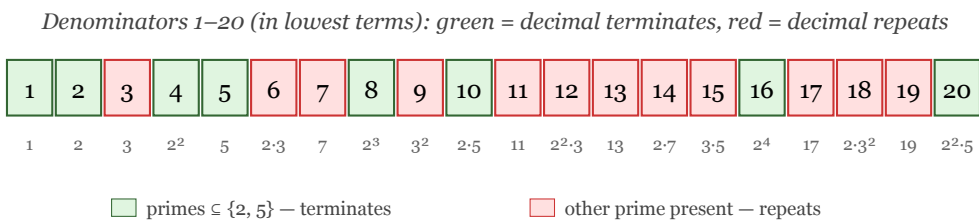


Figure 1.1: Each box shows a denominator b along with its prime factorization. Green boxes ($b = 1, 2, 4, 5, 8, 10, 16, 20$) have only 2s and 5s in their factorization — fractions a/b terminate. Red boxes have at least one other prime — fractions a/b repeat.

The eight green denominators in this strip are precisely the values of $2^m \cdot 5^n$ up to 20, namely $\{1, 2, 4, 5, 8, 10, 16, 20\}$. The set continues forever: $25, 32, 40, 50, 64, 80, 100, 125, \dots$ — dense in $\mathbb{Q}$ but countable.

1.5 Examples Side-by-Side

Terminates	Decimal	Doesn't terminate	Decimal
$1/2$	0.5	$1/3$	$0.\overline{3}$

$1/4$	0.25	$1/6$	$0.1\overline{6}$
$1/5$	0.2	$1/7$	$0.14285\overline{7}$
$1/8$	0.125	$1/9$	$0.\overline{1}$
$1/10$	0.1	$1/11$	$0.0\overline{9}$
$3/16$	0.1875	$5/12$	$0.41\overline{6}$
$7/20$	0.35	$22/7$	$3.14285\overline{7}$
$1/25$	0.04	$1/14$	$0.071428\overline{5}$
$1/40$	0.025	$1/15$	$0.0\overline{6}$
$17/125$	0.136	$8/21$	$0.38095\overline{2}$

Table 1.1

Every denominator on the left side factors as $2^m 5^n$; every denominator on the right has a prime factor outside $\{2, 5\}$.

1.6 Remark — Why the Connection to Base 10

Decimal expansions are *base-10* expansions, and $10 = 2 \cdot 5$. A fraction a/b has a finite expansion in base B if and only if (in lowest terms) the prime factors of b are all factors of B :

Base	Terminating denominators (in lowest terms)
$10 = 2 \cdot 5$	$b = 2^m 5^n$
2	$b = 2^m$ (i.e. powers of 2)
3	$b = 3^m$
$6 = 2 \cdot 3$	$b = 2^m 3^n$
$12 = 2^2 \cdot 3$	$b = 2^m 3^n$ (same set as base 6 — same prime support)

Table 1.2

So whether a fraction's expansion terminates is *not* an intrinsic property of the number — it depends on the choice of base. $1/3$ terminates in base 3 ($= 0.1_3$), even though it loops forever in base 10.

1.7 Remark — The Dual "0.999..." Representation

Every terminating decimal has a *second* valid representation ending in repeating 9s. For example:

$$0.5 = 0.4999\dots = 0.4\overline{9},$$

$$0.25 = 0.24999\dots = 0.24\overline{9}.$$

So the “terminating” form is not actually unique — but the underlying *number* is. A more honest characterisation: the set in question is the set of rationals admitting *at least one* terminating decimal expansion. (And those are exactly the rationals admitting at least one $\overline{9}$ -ending expansion too.)

1.8 Remark — Subring Structure

The set T of terminating decimals is closed under addition, subtraction, and multiplication, and contains 1. (Try it: $\frac{a}{2^m 5^n} + \frac{c}{2^p 5^q}$ has denominator dividing $2^{\max(m,p)} 5^{\max(n,q)}$.) Hence T is a *subring* of $\mathbb{Q}$, containing $\mathbb{Z}$:

$$\mathbb{Z} \subset T = \mathbb{Z}[1/10] \subset \mathbb{Q}.$$

It is *not* a field, however — for example $1/3 \notin T$, so not every nonzero element of T has a multiplicative inverse in T . (Only those with denominator a power of 10 actually have inverses also of the same form.)

The ring $\mathbb{Z}[1/10]$ is the *localisation* of $\mathbb{Z}$ at the multiplicative set $\{1, 10, 100, \dots\}$ — algebraists often write it as $\mathbb{Z}[\frac{1}{10}]$ or sometimes $\mathbb{Z}_{(10)}$ (the latter with a slightly different meaning in number-theory contexts; conventions vary).

1.9 Remark — Connection to Long Division

Here's an algorithmic way to see the same fact. Long-dividing a by b :

1. At each step you have a remainder $r \in \{0, 1, \dots, b-1\}$.
2. The next remainder is $(10r) \bmod b$, and the next decimal digit is $\lfloor 10r/b \rfloor$.
3. The expansion terminates iff some remainder eventually equals 0.

The remainder map $r \mapsto 10r \bmod b$ is *invertible* on the subgroup $\langle 10 \rangle$ of $(\mathbb{Z}/b\mathbb{Z})^\times$ if $\gcd(10, b) = 1$. If $\gcd(10, b) = 1$, the iterations cycle through invertible residues — *never* hitting 0. So no terminating expansion.

If $\gcd(10, b) > 1$ — i.e. b shares a prime factor with $10 = 2 \cdot 5$ — the iteration shrinks the "2-part and 5-part" of the remainder until b contributes no factors beyond 2, 5. At that point one further iteration kills the remainder.

The clean conclusion: termination is equivalent to b 's prime factors *all* dividing 10. This recovers our characterisation.

1.10 Remark — Cardinality

The set T of terminating decimals is **countable** — being a subset of $\mathbb{Q}$, which is countable. In fact T is *bijective* with $\mathbb{Z} \times \mathbb{Z}_{\geq 0}$ via the map $(c, k) \mapsto c/10^k$, which gives an explicit enumeration.

Despite being countable, T is *dense* in $\mathbb{R}$: between any two real numbers there is a terminating decimal. (Just truncate the decimal expansion of one of them at a sufficiently high place.) So T is countable + dense — a useful combination that makes terminating decimals an effective *practical* number system, even though they're a small slice of $\mathbb{Q}$.